

The data on the disk can be roughly divided into 5 parts according to their different characteristics and functions: MBR area, DBR area, FAT area, DIR area and DATA area. Let's introduce them separately:

(1) MBR area (main boot sector) MBR (Main Boot Record), according to its literal understanding, is the main boot record area, located in the entire hard disk 0 track 0 cylindrical 1 sector. However, out of a total of 512 bytes of the main boot sector, the MBR occupies only 446 of them (offset 0 - offset 1BDH), the remaining 64 bytes (offset 1BEH - offset 1FDH) are given to DPT (Disk Partition Table) (see table below), and the last two bytes "55, AA" (offset 1FEH - offset 1FFH) are the end mark of the partition. This whole constitutes the main boot sector of the hard drive. The master boot record contains a series of parameters and a bootloader for the hard disk. The main role of the hard disk bootloader is to check whether the partition table is correct, boot the operating system on the partition with the activation flag after the system hardware completes the self-test, and give control to the booter. MBR is generated by partitioning programs such as Fdisk.com, which do not depend on any operating system, and the hard drive bootloader can be changed, allowing for multi-system coexistence. Offset Length Meaning expressed 0 bytes Partition state: e.g., 0--> inactive partition

80--> Activity Division

1 byte The partition starts with the HEAD (HEAD) 2 words The partition starts sector and starts cylinder 4 bytes The partition type: e.g. 82--> Linux Native partition

83--> Linux Swap partition

5 bytes The partition termination HEAD (HEAD) 6 words The partition termination sector and the termination cylinder 8 double words The partition starts absolute partition C double word The number of partition sectors In the following sections, we will use an example to give you a more intuitive understanding of the main boot record:

Example: 80 01 01 00 0B FE FC 3F 00 00 00 7E 86 BB 00 Here we can see that the first "80" is an activation flag of a partition, indicating that the system is bootable; "01 01 00" means that the head number at the beginning of the partition is 01, the sector number at the beginning is 01, and the cylindrical number at the beginning is 00; "0B" indicates that the system type of the partition is FAT32, and the other more commonly used ones are 04 (FAT16) and 07 (NTFS). "FE FC" indicates that the head number at the end of the partition is 254, the sector number at the end of the partition is 63, and the cylindrical number at the end of the partition is 764; "3F 00 00 00" means that the relative sector number of the first sector is 63; "7E 86 BB 00" indicates a total sector count of 12289622.

(2) DBR area DBR (Dos Boot Record) is the meaning of the operating system boot record area. It is usually located in the 0 track 1 cylinder 1 sector of the hard disk and is the first sector that the operating system can directly access, and it includes a bootloader and a table of recording parameters for this partition called BPB (BiosParameter Block). The main task of the bootloader is to determine whether the first two files of this partition and directory are the boot files of the operating system (in the case of DOS, i.e. Io.sys and Msdos.sys) when the MBR gives it control of the system. If it is sure to exist, it is read into memory and the control is given to the file. The BPB parameter block records the start and end sectors, file storage format, hard disk media descriptor, root directory size, number of FAT, size of allocation units, and other important parameters of this partition.

(3) FAT area

After DBR is the more familiar FAT (File Allocation Table) area. Before explaining the concept of a file allocation table, let's talk about the concept of clusters. When a file takes up disk space, the base unit is not bytes but clusters. The size of the cluster is related to the specifications of the disk, generally speaking, each cluster of floppy disks is 1 sector, and the number of sectors per cluster of hard disks is related to the total capacity of the hard disk, which may be 4, 8, 16, 32, 64..... As we already know from the above, the data of the same file is not necessarily stored in its entirety in a contiguous area of the disk, but is often divided into segments like a chain. This storage method is called chain storage of files. Files on the hard drive often need to be created, deleted, grown, shortened, etc. The more you do this, the more fragmented the files on the disk may be (at least 1 cluster per segment). However, because the hard disk holds segment-to-segment connection information (i.e., FAT), the operating system is always able to accurately locate each segment and read it correctly when reading the file. However, this cluster-based storage method also has its flaws. This is mainly manifested in the use of space. The last cluster of each file may have space that is not fully utilized (called tail cluster space). Generally speaking, when the number of files is relatively large, each file wastes half a cluster of space on average. Well, let's get down to business, in order to achieve chain storage of files, it is necessary to accurately record which clusters have been occupied by files on the hard disk, and the cluster number of the next cluster that stores successor content must be specified for each occupied cluster, and for the last cluster of a file, it must be specified that this cluster has no successor cluster. These are held by FAT tables, which have many table entries, each recording a cluster of information. Because of the importance of FAT for file management, FAT has a backup, i.e. building another FAT of the same FAT behind the original FAT. All items in the initially formed FAT are marked as "unoccupied", but if the disk is partially damaged, the formatter detects the corrupted cluster and marks it as a "bad cluster" in the corresponding item, which will no longer be used when saving files in the future. The number of items in FAT is equivalent to the total number of clusters on the hard disk, and the number of bytes occupied by each item should also be corresponding to the total number of clusters, because the cluster number needs to be stored in it. There are many formats of FAT, the most common and familiar to readers are FAT16 and FAT32, where FAT16 refers to the use of 16-digit numbers in the file allocation table, because the 16-bit allocation table can manage up to 65536 clusters (i.e. 2 to the power of 16), that is, a specified hard disk partition. Since the maximum storage space per cluster is only 32KB, the maximum storage capacity per partition is only (65536×32 KB) when using FAT16 to manage hard drives, which is 2048MB, which is commonly referred to as 2G. For this reason, Microsoft has been using the FAT32 standard since Windows 95 OSR2 version, that is, using a 32-bit file allocation table to manage hard disk files, so that the system can assign up to 4294967296 (that is, 2 to the power of 32) clusters, so the same is true in the cluster At 32KB, the maximum capacity of each partition can reach more than 65G. In addition, when using FAT32 to manage hard drives, the cluster length in each disk is much smaller than that of the equivalent capacity disks managed using the FAT16 standard. Since the disk space occupied by files stored on the hard disk is the smallest unit of clusters, a file must occupy the entire cluster even if it is only a few tens of bytes. Therefore, FAT32 is more suitable for large hard drives.

(4) DIR area (root directory area) DIR (Directory) is the root directory area, immediately after the second FAT table (i.e., the backed up FAT table), which records the starting unit of each file (directory) in the root directory, the attributes of the file, etc. When locating the file location, the operating system can know the specific location and size of the file in the hard disk based on the starting unit in the DIR combined with the FAT table.

(5) DATA area The data area is the real data storage place, located behind the DIR area, occupying most of the data space on the hard disk.

FAT16 file system

The boot sectorA boot sector can be found in the first sector of every logical disk. On a floppy disk, the logicaldisk takes up all of the physical disk and the boot sector lives in the first physical sector. On a harddisk, a boot sector lives at the start of each partition. An example boot sec0000 EB 3E 90 4D 53 57 49 4E-34 2E 30 00 02 20 01 00 .>. MSWIN4.0.. .. 0010 02 40 03 00 00 F8 FF 00-3F 00 40 00 3F 00 00 00 .@.....?. @.? ... 0020 41 DE 1F 00 80 00 29 37-4B 84 32 46 55 4A 49 54 A....)7K.2FUJIT0030 53 55 31 32 32 34 46 41-54 31 36 20 20 20 F1 7D SU1224FAT16 .}0040 FA 33 C9 8E D1 BC FC 7B-16 07 BD 78 00 C5 76 00 .3.... {... x..v.0050 1E 56 16 55 BF 22 05 89-7E 00 89 4E 02 B1 0B FC . V.U.".. ^.. N...0060 F3 A4 06 1F BD 00 7C C6-45 FE 0F 8B 46 18 88 45|. E... F.. E0070 F9 FB 38 66 24 7C 04 CD-13 72 3C 8A 46 10 98 F7 .. 8f\$|... r<.F...0080 66 16 03 46 1C 13 56 1E-03 46 0E 13 D1 50 52 89 f..F..V..F...PR.0090 46 FC 89 56 FE B8 20 00-8B 76 11 F7 E6 8B 5E 0B F..V.. ..v...^..00A0 03 C3 48 F7 F3 01 46 FC-11 4E FE 5A 58 BB 00 07 ..H...F..N.ZX...00B0 8B FB B1 01 E8 94 00 72-47 38 2D 74 19 B1 0B 56rG8-t...V00C0 8B 76 3E F3 A6 5E 74 4A-4E 74 0B 03 F9 83 C7 15 .v>.. ^tJNt..... 00D0 3B FB 72 E5 EB D7 2B C9-B8 D8 7D 87 46 3E 3C D8 ;. r... +...}. F><.00E0 75 99 BE 80 7D AC 98 03-F0 AC 84 C0 74 17 3C FF u...}.....t. <.00F0 74 09 B4 0E BB 07 00 CD-10 EB EE BE 83 7D EB E5 t.....}.. 0100 BE 81 7D EB E0 33 C0 CD-16 5E 1F 8F 04 8F 44 02 ..}. 3...^....D.0110 CD 19 BE 82 7D 8B 7D 0F-83 FF 02 72 C8 8B C7 48 ...}. }....r... H0120 48 8A 4E OD F7 E1 03 46-FC 13 56 FE BB 00 07 53 H.N...F.. V...S0130 B1 04 E8 16 00 5B 72 C8-81 3F 4D 5A 75 A7 81 BF ...r..? Mzu... 0140 00 02 42 4A 75 9F EA 00-02 70 00 50 52 51 91 92 .. Bju...p.PRQ.. 0150 33 D2 F7 76 18 91 F7 76-18 42 87 CA F7 76 1A 8A 3..v... v.B... v..0160 F2 8A 56 24 8A E8 D0 CC-D0 CC 0A CC B8 01 02 CD .. V\$.0170 13 59 5A 58 72 09 40 75-01 42 03 5E 0B E2 CC C3 . YZXr.@u.B.^...0180 03 18 01 27 OD 0A 49 6E-76 61 6C 69 64 20 73 79 ...'...'.. Invalid sy0190 73 74 65 6D 20 64 69 73-6B FF OD 0A 44 69 73 6B stem disk... Disk01A0 20 49 2F 4F 20 65 72 72-6F 72 FF OD 0A 52 65 70 I/O error... Rep01B0 6C 61 63 65 20 74 68 65-20 64 69 73 6B 2C 20 61 lace the disk, a01C0 6E 64 20 74 68 65 6E 20-70 72 65 73 73 20 61 6E nd then press an01D0 79 20 6B 65 79 OD 0A 00-49 4F 20 20 20 20 20 20 y key... I001E0 53 59 53 4D 53 44 4F 53-20 20 20 53 59 53 80 01 SYSMSDOS SYS.. 01F0 00 57 49 4E 42 4F 4F 54-20 53 59 53 00 00 55 AA . WINBOOT SYS.. U.

Structure of the boot
sectoroffsetsizemeanig003jum
pinstruction038system
name0b33BIOS parameter
block3ec0bootstrap code

fe 2 sector signature

Structure of the BIOS paramter block

(BPB)offsetlengthmeaning000x2logical bytes per
sector020x1sectors per cluster030x2number of
reserved sectors starting from 0050x1number of
FATs060x2number of root dir entries080x2total
sectors. 0 if ofs 15h used0a0x1media
descriptor0b0x2number of sectors per
FAT0d0x2number of logical sectors per
track0f0x2number of logical heads110x4number
of hidden sectors150x4total
sectors190x1physical drive
number1a0x1reserved1b0x1signature byte for
extended boot record1c0x4serial
number200xblabel2b0x8FAT typeValues for the
example boot

sectoroffsetlengthmeaning000x2logical bytes
per sector020x1sectors per cluster030x2number
of reserved sectors starting from 0050x1number
of FATs060x2number of root dir
entries080x2total sectors. 0 if ofs 15h
used0a0x1media descriptor0b0x2number of
sectors per FAT0d0x2number of logical sectors
per track0f0x2number of logical
heads110x4number of hidden sectors150x4total
sectors190x1physical drive
number1a0x1reserved1b0x1signature byte for
extended boot record1c0x4serial
number200xblabel2b0x8FAT typeCheck the C
source code to define the FAT boot sector.

Structure of a FAT

disk disk address length contents

0 1 MBR

1 PhysSectPerTrck - 1 Undefined

PhysSectPerTrck 1 Boot sector

PhysSectPerTrck+1 NumFATs*NumSectPerFAT FATs

BookKeep-SectRootDir SectRootDir Root directory

BookKeep LogcDiskSize-BookKeep Data space

SectRootDir = RootDirSize / 16BookKeep = PhysSectPerTrck + 1 +

NumFATs * NumSectPerFAT + SectRootDirFAT

The FAT is an array of up to 65,536 16-bit unsigned integers. The first 2 and last 16 entries in theFAT are reserved. All other elements are of type: index into the FAT array. Each of thenon-reserved FAT entries correspond to a cluster on the disk. FAT indexMeaning as indexMeaning as value0x0000reservedcluster is available0x0001reservedundefined0x0002-0xffefdata clusternext cluster in chain is at this index0xffff0-0xffff6reservedundefined0xffff7reservedbad cluster0xffff8-0xfffffreservedend of the current cluster chainFAT entries 0x0000 and 0x0001 contain other information, usually 0xffff8 and 0xffff respectively. The fragment below is from the start of a FATFFF8 FFFF 0000 0004 FFFF 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000The fragment above shows a file using clusters 0x0003 & 0x0004. FoldersA folder is a data structure stored on the disk. The root folder is stored before the data area. Otherfolders are stored in the data area in the same way as files; These folders may grow in size, shrinkin size and be stored non-contiguously in the same way as files. The structure of all folders is as anarray of folder entries. The end of the folder is marked by a folder entry starting with a 00h byte. Folder entry structureoffsetlengthcontents0x000x8name0x080x3extension0x0b0x1flags0x0c0xarese rved0x160x2time0x180x2date0x1a0x2first cluter0x1c0x4sizeThe date field is in DOS date format, and the time field is in DOS time.

